

Your grade: 90%

Your latest: 90%

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To pass you need at least 80%. We keep your highest score.

Next item →

1.

What is the notation that we use to denote the pdf of the entire state trajectory $\{x_0, x_1, \dots, x_k\}$ conditioned on all measurements up to and including those at time k ?

1 / 1 point

☒

$f(x_k \mid Z_k).$

☐

$\hat{f}(x_k \mid Z_k).$

☐

$f(x_k \mid Z_k).$

☐

$\hat{f}(x_k \mid Z_k).$

☒ Correct

Yes, This is correct.
2.

When implementing the particle-filter sequential-importance-sampling (SIS) algorithm, how do we find $f(x_k^{(i)} \mid x_{k-1}^{(i)})$?

1 / 1 point

☒

From the process model.

☐

From the measurement model.

☐

It is a design choice. We can choose whatever is most convenient.

☐

We use a zero-mean Gaussian pdf having covariance matrix equal to the identity matrix.

☒ Correct

Yes, this is correct.
3.

When implementing the particle-filter sequential-importance-sampling (SIS) algorithm, how do we find $f(z_k \mid x_k^{(i)})$?

1 / 1 point

☐

From the process model.

☒

From the measurement model.

☐

It is a design choice. We can choose whatever is most convenient.

☐

We use a zero-mean Gaussian pdf having covariance matrix equal to the identity matrix.

☒ Correct

Yes, this is correct.
4.

When implementing the particle-filter sequential-importance-sampling (SIS) algorithm, how do we find $q(x_k^{(i)} \mid x_{k-1}^{(i)}, z_k)$?

1 / 1 point

☐

From the process model.

☐

From the measurement model.

☒

It is a design choice. We can choose whatever is most convenient.

☐

We use a zero-mean Gaussian pdf having covariance matrix equal to the identity matrix.

☒ Correct

Yes, this is correct.
5.

When we implement the SIS algorithm without resampling, to what value will the minimum weight converge?

1 / 1 point

0

☒ Correct

Yes, that is correct.
6.

When we implement the SIS algorithm without resampling, to what value will the maximum weight converge?

1 / 1 point

1

☒ Correct

Yes, that is correct.
7.

When we implement the SIS algorithm without resampling, to what value will the effective number of particles (effective sample size) converge?

1 / 1 point

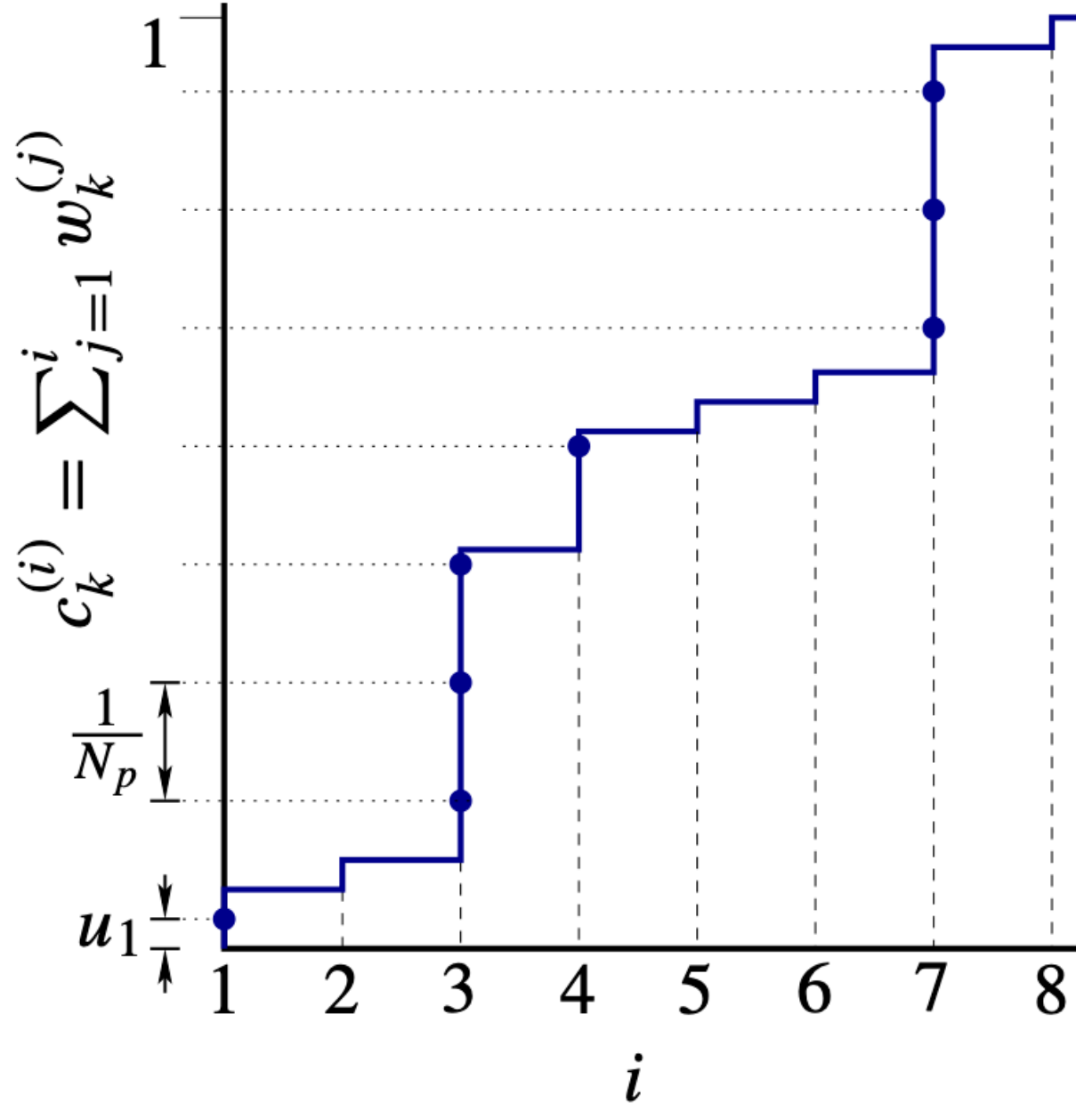
1

☒ Correct

Yes, that is correct.
8.

Consider the figure below, which is reproduced from slide 7 of Lesson 4.3.5. How many times is the fourth particle reproduced in the new population of particles after resampling?

1 / 1 point



1

☒ Correct

Yes, that is correct.
9.

Execute the Coursera Labs Jupyter notebook for Lesson 4.3.6. Set the number of particles to 600 and the resampling threshold to 300.

What is the root-mean-squared error between the estimate produced by the particle filter and the true state? Enter your answer with three digits to the right of the decimal point.

1 / 1 point

0.604

☒ Correct

Yes, that is correct.
10.

Execute the Coursera Labs Jupyter notebook for Lesson 4.3.6. Set the number of particles to 600 and the resampling threshold to 300.

What is the minimum number of effective particles stored by the SIS algorithm during the execution of the simulation? Enter your result with two digits to the right of the decimal point.

1 point

75.65

☒ Incorrect

No, that is not correct. Please confer Lesson 4.3.6.